Lecture -3
Supervised Learning

Supervised Learning

- Learning a discrete function: **Classification**
- Learning a continuous function: **Regression**

TYPES OF SUPERVISED LEARNING

Classification: To predict the outcome of a given sample where the output variable is in the form of categories (discrete values). Examples include labels such as male and female, sick and healthy.

Regression: To predict the outcome of a given sample where the output variable is in the form of real values (continuous values). Examples include real-valued labels denoting the amount of rainfall, the height of a person.

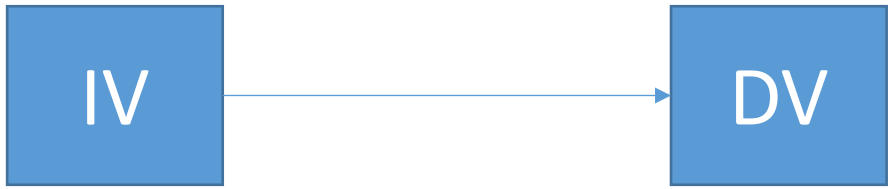
Regression

<u>Regression is divided into types:</u>

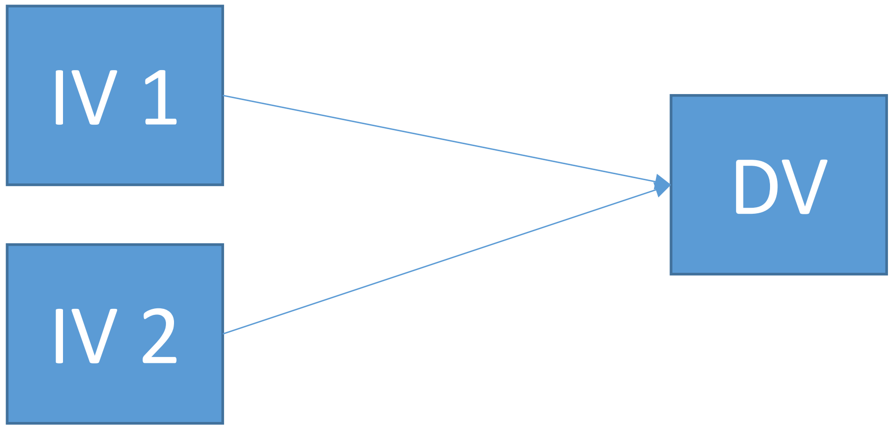
1. **Linear Regression**: Linear regression predictions are continuous values (rainfall in cm, predicting prices etc.)
2. **Logistic Regression**: Logistic regression predictions are discrete values (whether a student passed/failed) after applying a transformation function.

REGRESSION

- Simple Regression:
  - One Independent variable with one dependent variable



- Multiple Regression:
  - Two or more Independent variables with one dependent variable



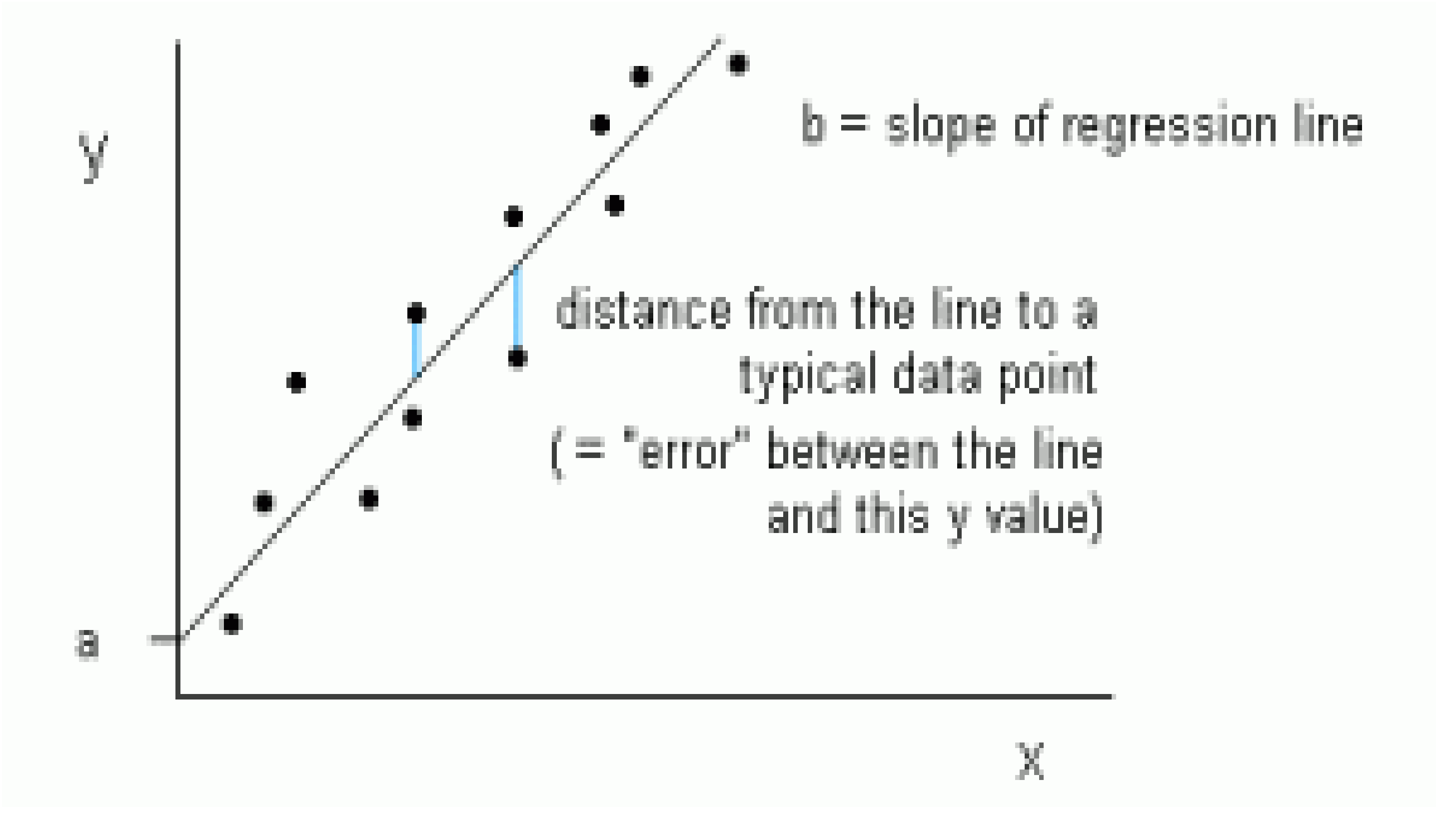
LINEAR REGRESSION

- Regression is a statistical method that helps us understand and predict relationship between variables.
- Describe how one variable (dependent variable) changes as another variable (independent variable) changes.
- Dependent variable: We are trying to predict or explain(Y).
- Independent variable: that are used to predict or explain the changes in the dependent variable(X).

For example: predicting salary based on years of experience, predicting exam score based on study, predicting resale based on vehicle sale.

Linear Regression

Linear regression is represented as a line in the form of y = a + bx.



Linear regression

Equation of Linear Regression: Y= mX+b

- Y represent the dependent variable
- X represent the independent variable
- m is the slope of the line(How much Y changes for a unit change in X)
- b is the intercept(the value of Y when X is 0)

Task: Predict Pizza Prices

- Step 1: Data Collection
- Step 2: Calculations
- Step 3: Prediction
- Step 4: Visualization

Example:

Pizza Diameter (x) (inches)	Price (y) (dollars)
8	10
10	13
12	16

What will be the price of pizza, if the diameter of pizza is 20 inches?

Solution:

Linear regression is represented as a line in the form of y = a + bx

$$a = \frac{[(\sum y)(\sum x^2) - (\sum x)(\sum xy)]}{[n(\sum x^2) - (\sum x)^2]}$$

$$b = \frac{[n(\sum xy) - (\sum x)(\sum y)]}{[n(\sum x^2) - (\sum x)^2]}$$

x	y	xy	x^2
8	10		
10	13		
12	16		

$$a = \frac{[(\sum y)(\sum x^2) - (\sum x)(\sum xy)]}{[n(\sum x^2) - (\sum x)^2]} \qquad =$$

$$b = \frac{[n(\sum xy) - (\sum x)(\sum y)]}{[n(\sum x^2) - (\sum x)^2]} \qquad =$$

$$a = \frac{[(\sum y)(\sum x^2) - (\sum x)(\sum xy)]}{[n(\sum x^2) - (\sum x)^2]} = -2$$

$$b = \frac{[n(\sum xy) - (\sum x)(\sum y)]}{[n(\sum x^2) - (\sum x)^2]} = 1.5$$

Now, predicting the price of pizza, if the diameter is 20 inches;

y = a + bx

y = -2 + (1.5) (20)

y = 28 dollars

Simple Linear Regression Code in Python

In [2]:
```python
import numpy as np
#This line imports the numpy library and renames it as np. numpy is a library for numerical computations in Python.
#It allows you to work with arrays and perform mathematical operations efficiently.
```

In [3]:
```python
from sklearn.linear_model import LinearRegression
#This line imports the LinearRegression class from
#the linear_model module of the scikit-learn library. scikit-learn
#is a popular machine learning library in Python, and LinearRegression is a class
#used for linear regression modeling.
```

In [4]:
```python
import matplotlib.pyplot as plt
#This line imports the matplotlib library for plotting data.
#You'll use it to create a graph to visualize your data and regression line.
```

In [5]:
```python
X = np.array([8, 10, 12]).reshape(-1, 1)
#This line creates a NumPy array X containing the independent variable (feature) data.
#It's a one-dimensional array that represents the values [8, 10, 12]. The reshape(-1, 1)
#part is used to reshape the array to have a single column, which is necessary for scikit-learn's LinearRegression class.
```

In [6]:
```python
y = np.array([10, 13, 16])
#This line creates a NumPy array y containing the dependent variable (target) data.
#It's also a one-dimensional array representing the corresponding target values [10, 13, 16].
```

In [7]:
```python
model = LinearRegression()
#This line creates an instance of the LinearRegression class and assigns it to the variable model.
#This object will be used for training and making predictions with the linear regression model.
```

In [8]:
```
model.fit(X, y)
#This line fits the linear regression model to the data. It trains the model to find the
#best-fitting linear relationship between the independent variable X and the dependent variable y.
```

In [9]:
```
X_new = np.array([20, 21, 22]).reshape(-1, 1)
#This line creates a new set of data points X_new for which you want to make predictions.
#It follows the same format as the original X, reshaped to a single column.
```

In [10]:
```
y_pred = model.predict(X_new)
#This line uses the trained model to predict the values of the dependent variable y for the new values in X_new.
#The predicted values are stored in the y_pred variable.
```

In [11]:
```
slope = model.coef_[0]
intercept = model.intercept_
#These lines calculate and store the slope (coefficient) and intercept of the linear regression line. The slope represents th
#where the line crosses the y-axis.
```

In [12]:
```
print(f"Slope (Coefficient): {slope}")
print(f"Intercept: {intercept}")
#These lines print out the calculated slope and intercept values to the console.
```

In [13]:
```
for i in range(len(X_new)):
    print(f"Prediction for X = {X_new[i][0]}: {y_pred[i]}")
#This loop prints the predictions for the new values in X_new to the console, showing
#the predicted values of the dependent variable for each corresponding new independent variable value.
```

In [14]:
```
plt.scatter(X, y, label='Data Points')
#This line creates a scatter plot of the original data points using matplotlib. It labels
#the data points as "Data Points" for the plot legend.
```

In [15]:
```
plt.plot(X_new, y_pred, color='red', label='Regression Line')
#This line plots the regression line (the line of best fit) using the predicted values from
#X_new. The line is displayed in red and labeled as "Regression Line" for the plot legend.
```

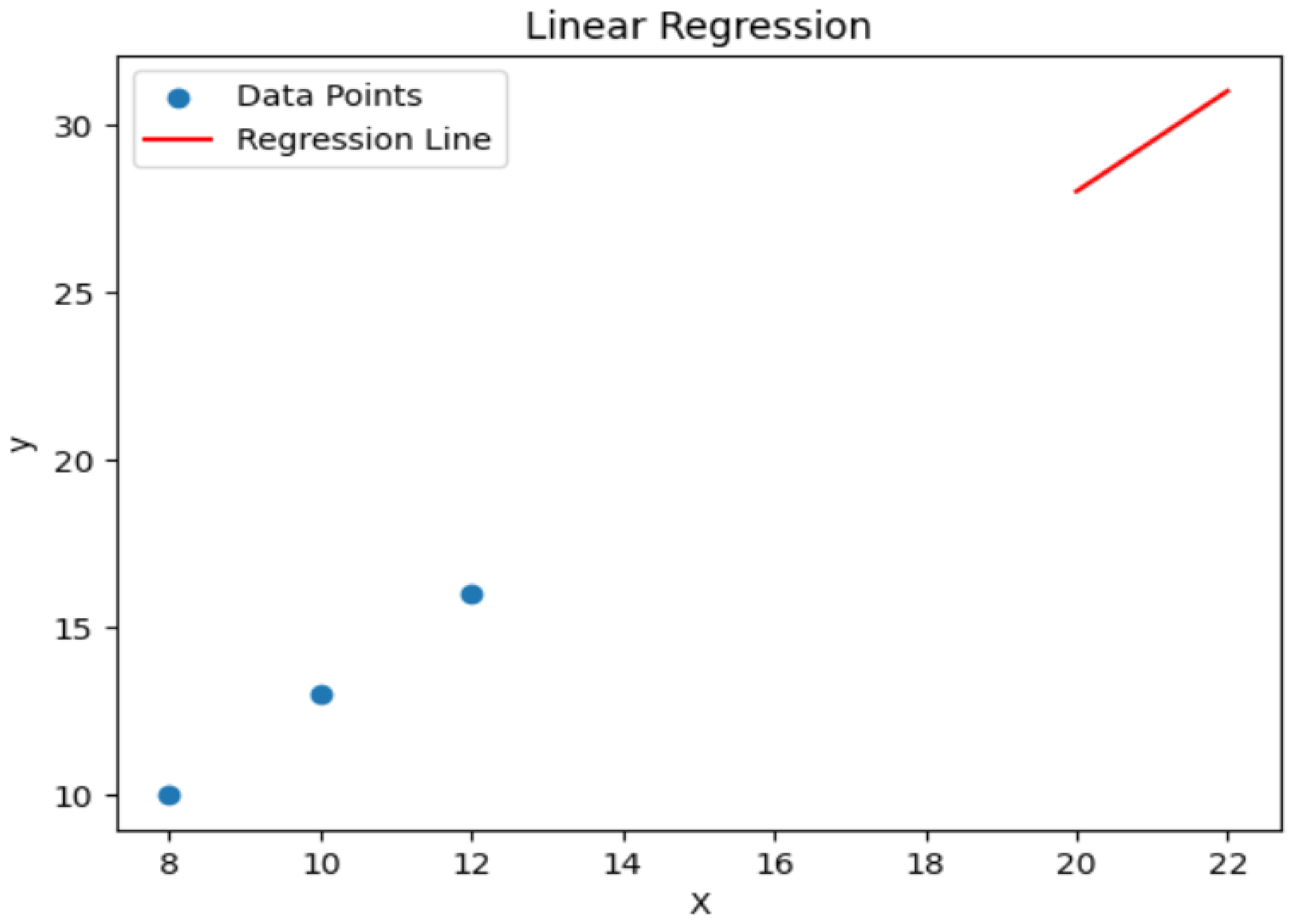
In [16]:
```
plt.xlabel('X')
plt.ylabel('y')
#These lines set the labels for the x-axis and y-axis on the plot.
```

In [17]:
```
plt.legend()
#This line adds a legend to the plot, which labels the data points and the regression line
```

In [18]:
```
plt.title('Linear Regression')
#This line sets a title for the plot.
```

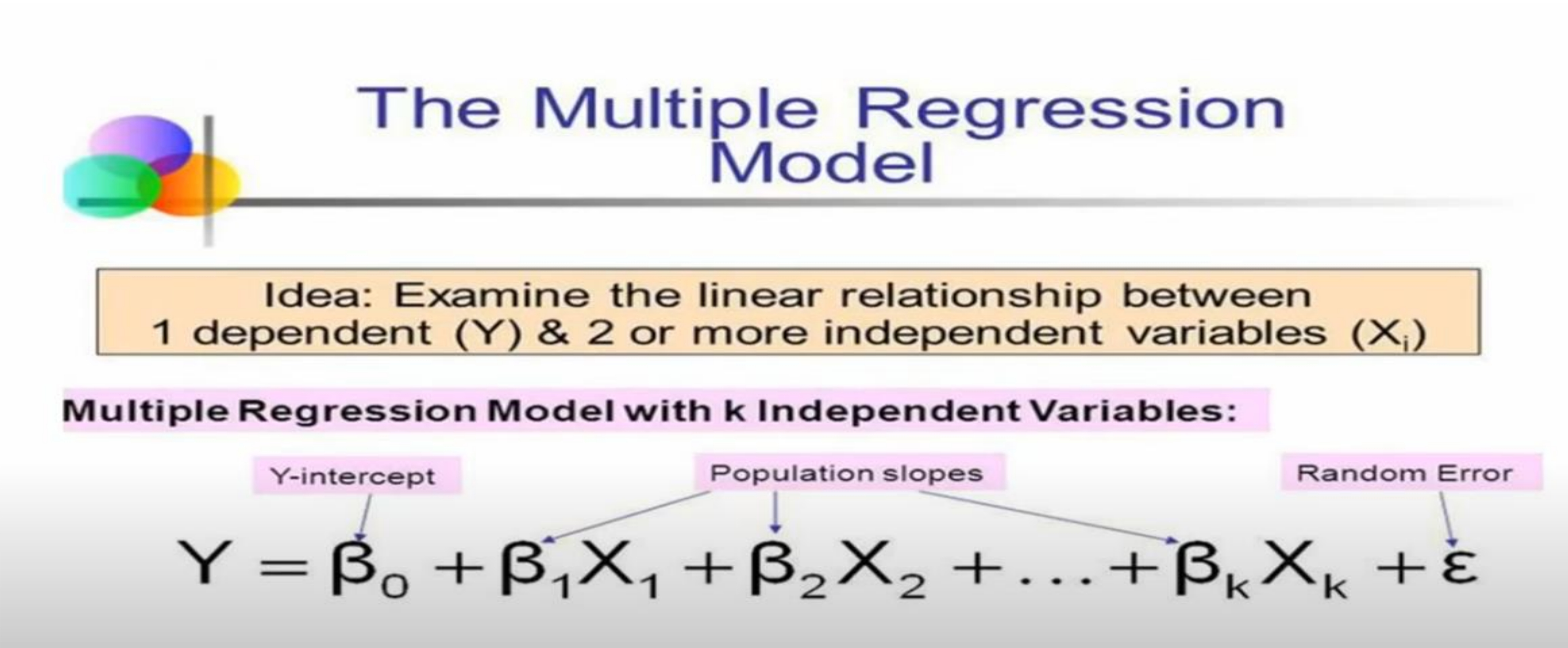
In [20]:
```
plt.show()
#This line displays the plot on the screen. When you run your script, the plot will appear in a graphical window.
```

Output of Python Program



Linear Regression Numerical Example with Multiple Independent Variable

Equation of Linear Regression: Y= mX+b



Predict the value of Y given X_1 and X_2

SUBJECT	Y	X_1	X_2
1	-3.7	3	8
2	3.5	4	5
3	2.5	5	7
4	11.5	6	3
5	5.7	2	1
6	?	3	2

Predict the value of Y given X1 and X2

- Finding the values of *b* (the slopes) is tricky for k>2 independent variables, and you really need matrix algebra to see the computation
- At this point, you should notice that all the terms from the one variable case appear in the two variable case.
- In the two variable case, the other X variable also appears in the equation. For example, X_2 appears in the equation for b_1. s.

Linear Regression 2 independent variable

$$a = \overline{Y} - b_1 \overline{X}_1 - b_2 \overline{X}_2$$

$$b_1 = \frac{(\sum x_2^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

$$b_2 = \frac{(\sum x_1^2)(\sum x_2 y) - (\sum x_1 x_2)(\sum x_1 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

Linear Regression 2 independent variable

$$\sum x_1^2 = \sum X_1 X_1 - \frac{(\sum X_1)(\sum X_1)}{N}$$

$$\sum x_2^2 = \sum X_2 X_2 - \frac{(\sum X_2)(\sum X_2)}{N}$$

$$\sum x_1 y = \sum X_1 Y - \frac{(\sum X_1)(\sum Y)}{N}$$

$$\sum x_2 y = \sum X_2 Y - \frac{(\sum X_2)(\sum Y)}{N}$$

$$\sum x_1 x_2 = \sum X_1 X_2 - \frac{(\sum X_1)(\sum X_2)}{N}$$

Linear Regression 2 independent variable

SUBJECT	Y	X_1	X_2	X_1X_1	X_2X_2	X_1X_2	X_1Y	X_2Y
1	-3.7	3	8	9	64	24	-11.1	-29.6
2	3.5	4	5	16	25	20	14	17.5
3	2.5	5	7	25	49	35	12.5	17.5
4	11.5	6	3	36	9	18	69	34.5
5	5.7	2	1	4	1	2	11.4	5.7

Linear Regression 2 independent variable

$$\sum x_1^2 = \sum X_1X_1 - \frac{(\sum X_1)(\sum X_1)}{N} = 10$$

$$\sum x_2^2 = \sum X_2X_2 - \frac{(\sum X_2)(\sum X_2)}{N} = 32.8$$

$$\sum x_1 y = \sum X_1Y - \frac{(\sum X_1)(\sum Y)}{N} = 17.8$$

$$\sum x_2 y = \sum X_2Y - \frac{(\sum X_2)(\sum Y)}{N} = -48$$

$$\sum x_1 x_2 = \sum X_1X_2 - \frac{(\sum X_1)(\sum X_2)}{N} = 3$$

SUBJECT	Y	X_1	X_2	X_1X_1	X_2X_2	X_1X_2	X_1Y	X_2Y
1	-3.7	3	8	9	64	24	-11.1	-29.6
2	3.5	4	5	16	25	20	14	17.5
3	2.5	5	7	25	49	35	12.5	17.5
4	11.5	6	3	36	9	18	69	34.5
5	5.7	2	1	4	1	2	11.4	5.7
$\sum$	19.5	20	24	90	148	99	95.8	45.6

Linear Regression 2 independent variable

$$b_1 = \frac{(\sum x_2^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2} = \frac{32.8 * 17.8 - 3 * (-48)}{10 * 32.8 - 3 * 3} = 2.28$$

$$b_2 = \frac{(\sum x_1^2)(\sum x_2 y) - (\sum x_1 x_2)(\sum x_1 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2} = \frac{10 * (-48) - 3 * 17.8}{10 * 32.8 - 3 * 3} = -1.67$$

$$a = \overline{Y} - b_1 \overline{X}_1 - b_2 \overline{X}_2 = \frac{19.5}{5} - \frac{2.28 * 20}{5} - \frac{-1.67 * 24}{5} = 2.796$$

Linear Regression 2 independent variable

Final Regression equation or Model is:

Y-intercept Population slopes Random Error

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k + \varepsilon$$

Final Regression equation or Model is:

$$\boldsymbol{Y = 2.796 + 2.28\,x_1 - 1.67x_2}$$

$$Now\ given \quad x_1 = 3 \quad and \quad x_2 = 2 \qquad Y = ?$$

$$Y = 2.796 + 2.28 * 3 - 1.67 * 2$$
$$= \mathbf{6.296}$$

Multiple Linear Regression Code in Python

Home WORK

```
# Sample data (you can replace this w
X1 = np.array([5,8,9,11,14])  # Firs
X2 = np.array([4,9,13,16,19])   # Sec
y = np.array([8.5,4.5,6.1,12.3,9.3])
```

```
# Make predictions for new data points
X1_new = np.array([4])
X2_new = np.array([6])
```

Answer

```
Regression Coefficients (for X1 and X2): [-0.4831746   0.49174603]
Intercept: 6.682539682539686
Prediction for X1 = 3, X2 = 2: 6.216507936507938
```

In [2]:

```python
# Import the necessary libraries
import numpy as np
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt  # Import Matplotlib for plotting

# Sample data (you can replace this with your own data)
X1 = np.array([3,4,5,6,2])  # First independent variable
X2 = np.array([8,5,7,3,1])   # Second independent variable
y = np.array([-3.7,3.5,2.5,11.5,5.7]) # The value we want to predict

# Combine the two independent variables into a single dataset
data = np.column_stack((X1, X2))

# Create a Linear Regression model
model = LinearRegression()

# Train (fit) the model using our data
model.fit(data, y)

# Make predictions for new data points
X1_new = np.array([3])
X2_new = np.array([2])
new_data = np.column_stack((X1_new, X2_new))
y_pred = model.predict(new_data)

# Print the results
print("Regression Coefficients (for X1 and X2):", model.coef_)
print("Intercept:", model.intercept_)

# Print the predictions for the new data points
for i in range(len(X1_new)):
    print(f"Prediction for X1 = {X1_new[i]}, X2 = {X2_new[i]}: {y_pred[i]}")
```

```
Regression Coefficients (for X1 and X2): [ 2.28163009 -1.67210031]
Intercept: 2.7995611285266437
Prediction for X1 = 3, X2 = 2: 6.300250783699057
```

Practice Problems

1- A company manufactures an electronic device to be used in a very wide temperature range. The company knows that increased temperature shortens the life time of the device, and a study is therefore performed in which the life time is determined as a function of temperature. The following data is found: What will be lifetime of device if temperature is 60 Celsius?

Temperature in Celsius	Lifetime in Hours
10	420
20	365
30	285
40	220

2- A company sales (in millions) for 5 years (2019 – 2023) are:

- X (years since 2019) : 0, 1, 2, 3, 4
- Y (Sales in millions) : 12, 19, 29, 37, 45

Predict sales in 2026.

3- A student wants to predict test scores based on hours studied:

- X (hours) : 2, 3, 4, 5, 6
- Y (score): 60, 65, 75, 80, 90

If a student studies for 8 hours, what is their predicted score.

4- A real estate model is developed to predict house prices based on Square foots (X1) and no. of bedrooms (X2). The output is:

Price = 50,000 + 150 (X1) + 20,000 (X2)

Predict the price of a 20,000 sq.ft house with 3 bedrooms.

5- A company wants to predict sales (Y, in 1000s) based on TV advertising budget (X1, in 1000s) and Radio advertising budget (X2, in 1000s):

- X1 : 2, 4, 6, 8, 10
- X2 : 1, 2, 3, 4, 5
- Y : 10, 15, 20, 25, 30

Predict sales for X1 = 12 and X2 = 6.

Logistic Regression

Logistic regression refers to any regression model in which the response variable is categorical.

There are three types of logistic regression models:

- **Binary logistic regression**: The response variable can only belong to one of two categories.
- **Multinomial logistic regression**: The response variable can belong to one of three or more categories and there is no natural ordering among the categories.
- **Ordinal logistic regression**: The response variable can belong to one of three or more categories and there *is* a natural ordering among the categories.

Example:

Suppose a sports data scientist wants to use the predictor variables (1) points, (2) rebounds, and (3) assists to predict the probability that a given college basketball player gets drafted into the Academy.

Since there are only two possible outcomes (drafted or not drafted) for the response variable, the data scientist would use a binomial logistic regression model.

Example:

Suppose a political scientist wants to use the predictor variables (1) annual income and (2) years of education to predict the probability that an individual will vote for one of four different presidential candidates.

Since there are more than two possible outcomes (there are four potential candidates) for the response variable and there is no natural ordering among the outcomes, the political scientist would use a multinomial logistic regression model.

Examples:

Suppose a business wants to use the predictor variables (1) word count and (2) country of origin to predict the probability that a given email is spam.

Since there are only two possible outcomes (spam or not spam) for the response variable, the business would use a binomial logistic regression model.

Examples:

Suppose an academic advisor wants to use the predictor variables (1) GPA, (2) Exam score, and (3) SAT Exam score to predict the probability that an individual will get into a university that can be categorized into “bad”, “average”, “good”, or “great.”

Since there are more than two possible outcomes (there are four classifications of school quality) for the response variable and there *is* a natural ordering among the outcomes, the academic advisor would use an ordinal logistic regression model.

Examples:

Suppose a sports analyst wants to use the predictor variables (1) TV hours viewed per week and (2) age to predict the probability that an individual will pick either basketball, football, or cricket as their preferred sport.

Since there are more than two possible outcomes (there are three sports) for the response variable, the sports analyst would use a multinomial logistic regression model.

Examples:

Suppose a movie critic wants to use the predictor variables (1) total run time and (2) genre to predict the probability that a given movie will receiving a rating between 1 and 10.

Since there are more than two possible outcomes (there are 10 possible ratings) for the response variable and there *is* a natural ordering among the outcomes, the movie critic would use an ordinal logistic regression model.

Logistic regression

- Supervised classification model
- Logistic regression is a statistical method used for binary classification tasks, where the outcome variable is categorical with two possible values, often coded as 0 and 1. It estimates the probability that a given input belongs to one of the two classes.
- The logistic function, also known as the sigmoid function, is a key component of logistic regression.
- Depending variable is categorical and binary (0 or 1)

Study Hours	Exam Result
2	0
3	0
4	0
6	1
7	1
8	1
5	**?**

Logistic regression can also be used in the following areas:

- In healthcare to identify risk factors for diseases and plan preventive measures.
- In drug research to tease apart the effectiveness of medicines on health outcomes across age, gender and ethnicity.
- In weather forecasting apps to predict snowfall and weather conditions.
- In political polls to determine if voters will vote for a particular candidate.
- In insurance to predict the chances that a policyholder will die before the policy's term expires based on specific criteria, such as gender, age and physical examination.
- In banking to predict the chances that a loan applicant will default on a loan or not, based on annual income, past defaults and past debts.

LOGISTIC REGRESSION- NUMERICAL EXAMPLE 01

Equation of Linear Regression: Y= mX+b

a0 = -1.5

a1 = 0.6

X= 5

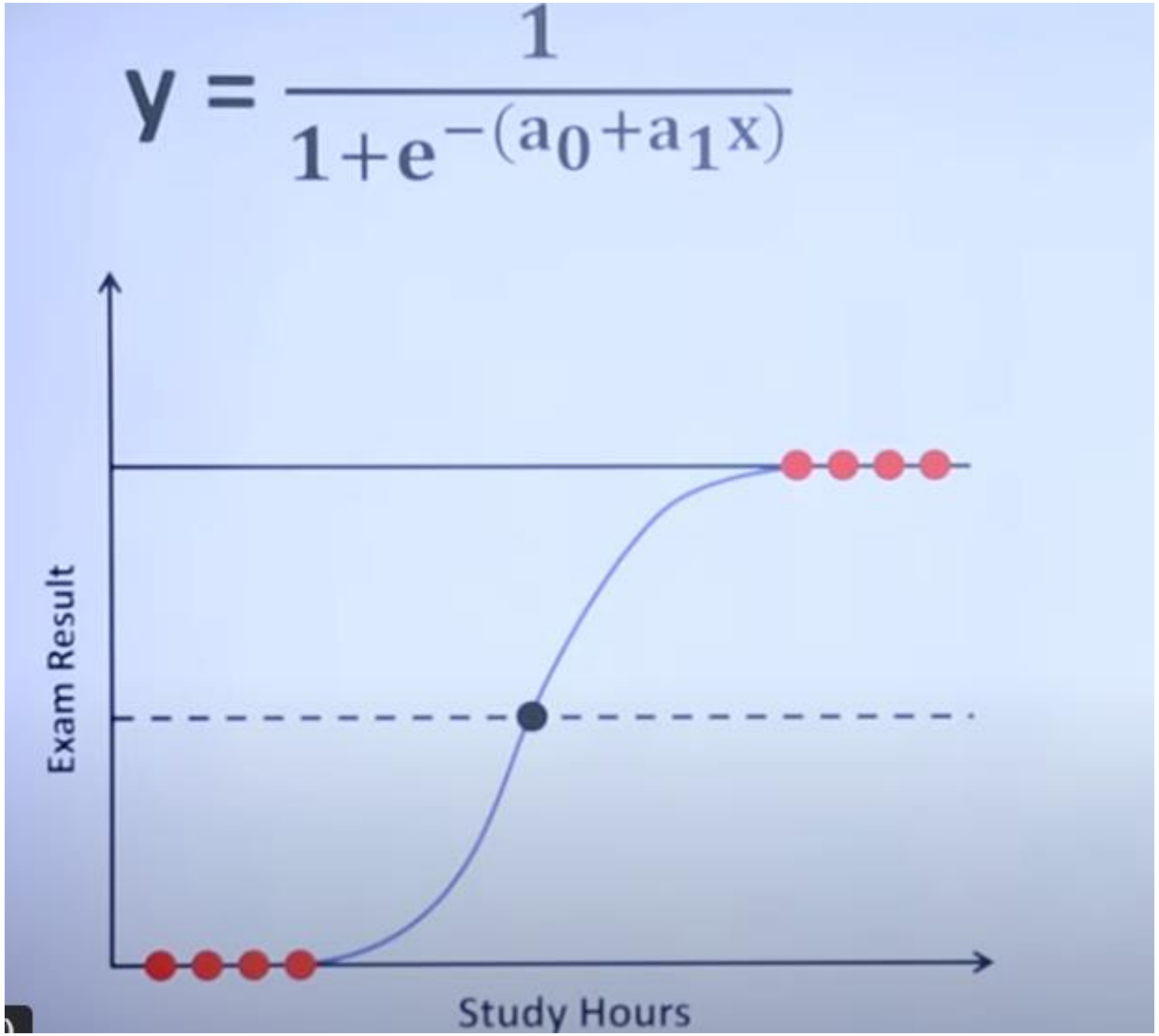
Note: - slope -> m -> a1

- intercept -> b -> a0

$$= 1 \div (1 + e^{-(-1.5 + 0.6 \times 5)})$$

The value of e is equal to 2.71828 or e ≈ 2.72.

$$= 0.81757447619$$



Logistic regression- multiple variables

Let’s say that we have a training set of 500 data

X1	X2	Y
2.1	2.5	0
1.4	2.3	0
3.3	4.4.	0
8.06	3.05	0
5.3	2.75	0
2.7	**2.5**	**?**

LOGISTIC REGRESSION- NUMERICAL EXAMPLE WITH MULTI VARIABLES

$$= \frac{1}{1 + e^{-(b_0 + b_1 * x_1 + b_2 \cdot x_2)}}$$

Initially we assume B0 = 0.0 ,B1 = 0.0 & B2= 0.0

Let's take X1 = 2.7, X2 = 2.5
Y = ?

$$Y = 1 \div (1 + e^{-(0.0 + 0.0 \times 2.7 + 0.0 \times 2.5)})$$

$$Y = 0.5$$